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CRITICALLY APPRAISED TOPIC

The Functional Movement Screen's Ability to Detect Changes in Movement Patterns After a Training Intervention

Lindsay M. Minthorn, Shirleeah D. Fayson, Lisa M. Stobierski, Cailee E. Welch, and Barton E. Anderson

Clinical Scenario: Appropriate movement patterns during sports and physical activities are important for both athletic performance and injury prevention. The assessment of movement dysfunction can assist clinicians in implementing appropriate rehabilitation programs after injury, as well as developing injury-prevention plans. No gold standard test exists for the evaluation of movement capacity; however, the Functional Movement Screen (FMS) has been recommended as a tool to screen for movement-pattern limitations and side-to-side movement asymmetries. Limited research has suggested that movement limitations and asymmetries may be linked to increased risk for injury. While this line of research is continuing to evolve, the use of the FMS to measure movement capacity and the development of intervention programs to improve movement patterns has become popular. Recently, additional research examining changes in movement patterns after standardized intervention programs has emerged. ***Clinical Question:*** Does an individualized training program improve movement patterns in adults who participate in high-intensity activities?

Keywords: asymmetrical movement, corrective exercises, intervention programs

Clinical Scenario

Appropriate movement patterns during sports and physical activities are important for both athletic performance and injury prevention. The assessment of movement dysfunction can assist clinicians in implementing appropriate rehabilitation programs after injury, as well as developing injury-prevention plans. No gold standard test exists for the evaluation of movement capacity; however, the Functional Movement Screen (FMS) has been recommended as a tool to screen for movement-pattern limitations and side-to-side movement asymmetries. Limited research has suggested that movement limitations and asymmetries may be linked to increased risk for injury. While this line of research is continuing to evolve, the use of the FMS to measure movement capacity and the development of intervention programs to improve movement patterns has become popular. Recently, additional research examining changes in movement patterns after standardized intervention programs has emerged.

The authors are with the Athletic Training Program, A.T. Still University, Mesa, AZ. Address correspondence to Cailee Welch at cwelch@atsu.edu.

Clinical Question

Does an individualized training program improve movement patterns in adults who participate in high-intensity activities?

Summary of Search, "Best Evidence" Appraised, and Key Findings

- The literature was searched for studies of level 3 evidence or higher that assessed movement patterns using the FMS before and after a training program.
- The literature search returned 3 possible studies[1–3] related to the clinical question; all 3 studies met the inclusion criteria and were included.
- One of the included studies[2] reported no significant change in movement patterns after a 12-week training program.
- The other 2 studies[1,3] reported significant changes in movement patterns after training programs.

Clinical Bottom Line

There is inconsistent evidence to demonstrate that individualized training programs can improve movement

patterns in adults who participate in high-intensity activities. This critically appraised topic (CAT) revealed conflicting findings in the current literature as to whether individualized training programs can improve movement patterns. Some evidence suggests that movement patterns improve after a training program,[1,3] while other evidence suggests that movement patterns may not improve after a training program.[2] Therefore, it remains unclear whether an individualized training program can improve movement patterns in adults who participate in high-intensity activities. Clinicians wishing to implement intervention programs to improve movement patterns should take into consideration these inconsistent results and consider using other outcome measures, in addition to the FMS, to determine the effectiveness of training interventions.

Strength of Recommendation: Due to the inconsistent results of the included studies, there is grade C evidence that an individualized training program can improve movement patterns in adults who participate in high-intensity activities.

Search Strategy

Terms Used to Guide Search Strategy

- **P**atient/Client group: *physically active individuals* or *athletes*
- **I**ntervention (or assessment): *individualized training program* or *intervention program*
- **C**omparison: *control group* (no individualized training program)
- **O**utcome(s): *movement patterns* and *Functional Movement* Screen or *FMS*

Sources of Evidence Searched

- The Cochrane Library
- Medline
- CINAHL
- SPORTDiscus
- Additional resources obtained via review of reference lists and hand search

Inclusion and Exclusion Criteria

Inclusion Criteria

- Level 3 evidence or higher
- Studies that assessed movement patterns, as measured by the FMS, before and after individualized training programs
- Studies that included participants in high-intensity activities
- Limited to English language
- Limited to the past 10 years (2004–2013)

Exclusion Criteria

- Studies that included participants younger than 18 years of age
- Studies that did not use the FMS to assess movement patterns
- Studies that included participants with current musculoskeletal injuries or restricted training

Results of Search

Three relevant studies[1–3] were located and categorized as shown in Table 1 (based on Levels of Evidence, Oxford Centre for Evidence Based Medicine, 2011).

Best Evidence

The studies in Table 2 were identified as the best evidence and selected for inclusion in this CAT. These studies were selected because they were considered level 3 evidence or higher and assessed movement patterns of individuals before and after individualized training programs.

Implications for Practice, Education, and Future Research

Based on this appraisal, 2 of the 3 included studies[1,3] indicated that an individualized training program can improve movement patterns in adults participating in high-intensity activities. However, 1 study[2] did not find significant differences in movement patterns after an individualized training program. The inconsistent findings from this appraisal may be due to differences in study design and methods or due to other limitations associated with the clinical utility of the FMS.

Two of the studies[1,2] used control groups, in which participants did not receive the individualized training program. Kiesel et al[3] used a cohort design where participants were stratified into 2 groups (football lineman, football non-lineman); each group completed the same exercise program. The lack of a control group represents a potential limitation of this study, making it

Table 1 Summary of Study Designs of Articles Retrieved

Level of evidence	Study design	Number located	Reference
1b	Randomized control trial (cluster)	1	Bodden et al[1]
1b	Randomized control trial (single blinded)	1	Frost et al[2]
2b	Individual cohort	1	Kiesel et al[3]

Table 2 Characteristics of Included Studies

	Bodden et al[1]	Frost et al[2]	Kiesel et al[3]
Study design	Randomized cluster	Randomized controlled trial	Individual cohort
Participants	25 male semiprofessional mixed martial arts (MMA) athletes (24.31 ± 4.46 y) free of musculoskeletal disorder and in full, unrestricted training. Participants were stratified into an intervention (n = 12) or control (n = 13) group based on their gym location. Participants instructed not to change any normal training routines.	60 male firefighters (37.5 ± 9.8 y) free of musculoskeletal disorder or pain and participating in full, unrestricted duty. Participants were assigned to 1 of 3 groups, intervention 1 (n = 21), intervention 2 (n = 19), or control group (n = 20), and groups were matched for age, height, body mass, and baseline FMS scores. Control group instructed not to change activity habits during the 12-wk period of the study.	62 healthy professional American football players participating in a structured off-season strength and conditioning program. Participants were grouped by position: lineman and linebackers (n = 32) and nonlinemen (n = 30)
Intervention investigated	Intervention group completed an 8-wk (4 sessions) training program. Intervention groups' training exercises were as recommended by the FMS advanced corrective exercise manual. The exercises programs were given to participants via video and paper file. Individual programs focused on the weakest and/or asymmetrical scores, with primary focus on mobility patterns and secondary moving on to stability patterns if appropriate. FMS scores were measured at baseline, wk 4, and wk 8 of the training program.	Intervention groups completed a12-wk (3 sessions) training program. Both intervention groups were coached by strength and conditioning professionals. Intervention 1 group's training focused on whole-body coordination and control during task execution. Strength, power, aerobic capacity, and injury prevention were priority. Intervention 2 group's training focused on maximizing performance and fitness outcomes alone. Participants required to complete at least 30/36 training sessions. FMS scores were measured at baseline and wk 12 of the training program.	All participants completed a 7-wk (4 sessions plus 2 optional) training program. Off-season strength and conditioning program, individualized based on pretest FMS scores designed to correct identified movement deficits. Movement preparation exercises included self-trigger point massage with the stick for major muscle groups, stride self-stretch, straight-leg partner stretch, and leg lowering. Corrective exercises included straight-leg bridge, single-leg stance with core engagement, leg lowering with core engagement, and single-leg toe touch with stick. Individualized corrective exercises were prescribed and preformed with the supervision and instruction of the strength and conditioning staff. FMS scores were measured at baseline and wk 7 of the training program.

(continued)

Table 2 *(continued)*

	Bodden et al[1]	Frost et al[2]	Kiesel et al[3]
Outcome measure	FMS scores	FMS scores	FMS scores
Main findings	No significant differences existed between groups at baseline ($P = .96$). Baseline scores were 13.23 ± 0.80 in the control group and 13.25 ± 0.87 in the intervention group. A significant time-by-group interaction effect for FMS test scores was noted ($P < .01$), with scores increasing between wk 0 and 4 ($P = .00$) and no significant increase between wk 4 and 8 ($P = 1.00$). Scores did not change across time in the control group.	No significant differences existed between groups at baseline for the standard scoring (control = 13.3 ± 2.5, intervention 1 = 13.1 ± 2.7, intervention 2= 12.8 ± 1.7). No significant ($P = .0176$) posttraining changes in the total FMS score for any group (control = 13.0 ± 2.4, intervention 1 = 13.5 ± 2.3, intervention 2 = 13.1 ± 1.8). The total number of asymmetries present before and after training was found to be just as variable as the individual screen scores, and there were no differences between groups ($P = .528$).	There was no time by position interaction for the FMS scores over the course of the training program ($P = 50.78$). The pretest score for the linemen was 11.8 ±1.8 and for the nonlinemen was 13.3 ±1.9. The mean posttest score for the lineman was 14.8 ± 2.4 and for the nonlinemen was 16.3 ± 2.4. 41 players were free from asymmetry at the end of the study compared with the 31 at the beginning of the study. There was a main effect of time ($P = .01$).
Level of evidence	1b	1b	2b
Validity score	PEDro score 6/10	PEDro score 5/10	N/A
Conclusion	A standardized exercise training intervention can improve dysfunctional movement patterns and increase FMS scores. These results cannot be generalized beyond the population tested, MMA fighters. Results may not be the same in a less elite population.	The average composite FMS scores did not change after 12 wk of training, and there were no differences among the 3 groups.	Off-season training program can significantly improve scores on the FMS. These results cannot be generalized beyond the population tested, professional football players. Results may not be the same in a nonelite population.

more difficult to attribute score improvements to the intervention versus repeated measures. Baseline FMS scores across all studies were similar for all groups (lowest 12.8 ± 1.7, highest 13.3 ± 1.9) except football linemen (11.8 ± 1.8), with no significant differences between control and intervention groups. This suggests that most groups possessed similar movement capacity preintervention, and thus similar capacity for improvement.

Two of the studies[1,3] reviewed in this CAT used the exercise algorithm outlined by Cook[4] for their individualized training programs. This algorithm uses specific corrective exercises performed 4 times per week, that target movement pattern limitations related to the 7 FMS tasks. The intervention programs used by Frost et al[2] focused on improving whole-body coordination and control (intervention 1) and maximizing performance and fitness (intervention 2) rather than focusing on specific movement-pattern limitations. These differences in intervention programs may help explain the inconsistent results found in this CAT.

Once a patient's movement-pattern limitations are identified, a clinician can develop an individualized training program to address the limitations. Based on the results of this review, intervention programs that include exercises that are directly related to specific movement patterns may be successful.[1,3] In contrast, intervention programs that focus on whole-body coordination and control or maximizing performance and fitness may not be successful in improving movement patterns.[2]

Previous research has identified a cutoff score of 14 on the FMS, which suggests that individuals who score less than 14 are at an increased risk for injury.[5,6] However, further research is necessary to determine if injury risk is reduced when an individual's movement-pattern limitations or asymmetries are resolved and FMS scores improve. Additional research should also explore the interpretation of FMS scores in relation to exercise prescription and long-term intervention to make certain that the training is consistent and effective. Furthermore, exploring the FMS in other populations may yield varying insights. For example, the study by Keisel et al[3] suggested that professional football players can improve their movement patterns with an off-season training program. However, it is unknown if similar results would be achieved in athletes participating in other sports, in a nonathletic population, or at other age and maturation levels. Finally, a recent study by Frost et al[7] demonstrated changes in FMS scores of fire fighters simply by providing participants with knowledge of the FMS grading criteria. They concluded that the FMS scores improved with the participants' increased awareness of the task. This information brings into question the ability of the FMS to measure change over time or to measure transfer of training to movement-pattern improvement, especially if the participants have knowledge of the FMS grading criteria. It also demonstrates the need for continued research to determine the clinical utility of the FMS.

The knowledge from the articles[1–3] included in this CAT presents inconsistent evidence that an individualized training program may improve movement-pattern limitations, as identified by the FMS. The FMS composite score offers an objective measurement of movement quality. Movement quality can be assessed before, throughout, and after an intervention program because the FMS is both time- and cost-efficient. Since FMS scoring is standardized, clinicians across varying professions can use this tool.[2] However, clinicians should be aware of recent findings that bring into question the ability of the FMS to measure movement capacity and accurately identify true dysfunctional movement patterns.[7] Those wishing to assess movement capacity should consider using the FMS as 1 component in a comprehensive functional evaluation. Clinicians should also be aware that little evidence currently exists to correlate FMS-score increases with improved athletic performance.[8] This CAT should be reviewed in 2 years or when additional best evidence becomes available to determine whether additional best evidence has been published that may change the clinical bottom line for the research question posed in this review.

References

1. Bodden JG, Needham RA, Chockalingham N. The effect of an intervention program on functional movement screen test scores in mixed martial arts athletes. *J Strength Cond Res*. 2015;29(1):219–225.
2. Frost DM, Beach TAC, Callaghan JP, McGill SM. Using the Functional Movement Screen to evaluate the effectiveness of training. *J Strength Cond Res*. 2012;26(6):1620–1630. PubMed
3. Kiesel K, Plisky P, Butler R. Functional movement screen test scores improve following a standardized off-season intervention program in professional football players. *Scand J Med Sci Sports*. 2011;21:287–292. PubMed doi:10.1111/j.1600-0838.2009.01038.x
4. Cook G. *Movement: Functional Movement Systems*. Ramsey, NJ: On Target Publications; 2010.
5. Kiesel K, Plisky PJ, Voight ML. Can serious injury in professional football be predicted by a preseason functional movement screen? *N Am J Sports Phys Ther*. 2007;2(3):147–158. PubMed
6. Chorba RS, Chorba DJ, Bouillon LE, et al. Use of a functional movement screening tool to determine injury risk in female collegiate athletes. *N Am J Sports Phys Ther*. 2010;5(2):47–54. PubMed
7. Frost DM, Beach TA, Callaghan JP, McGill SM. FMS scores change with performers' knowledge of the grading criteria—are general whole-body movement screens capturing "dysfunction" [published online ahead of print November 20, 2013]? *J Strength Cond Res*. PubMed
8. Okada T, Huxel KC, Nesser TW. Relationship between core stability, functional movement, and performance. *J Strength Cond Res*. 2011;25(1):252–261. PubMed doi:10.1519/JSC.0b013e3181b22b3e

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